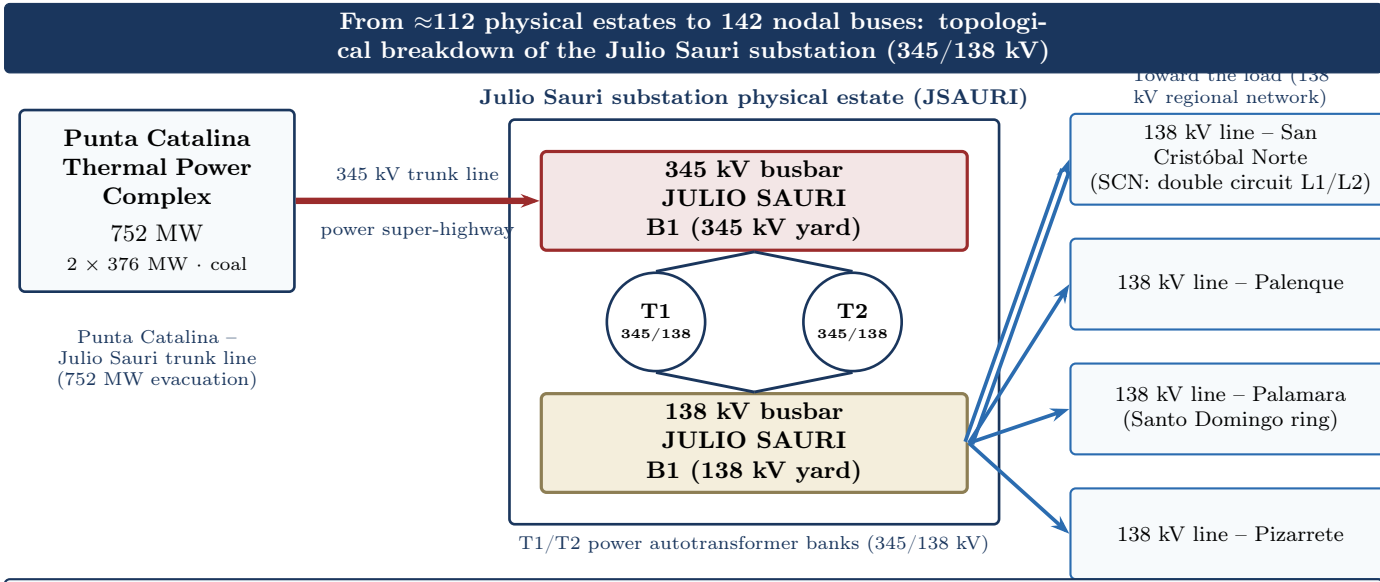

From ≈112 physical estates to 142 nodal buses: topological breakdown of the Julio Sauri substation (345/138 kV)
Julio Sauri substation physical estate (JSAURI)
Toward the load (138 kV regional network)
Punta Catalina Thermal Power Complex
752 MW
2 × 376 MW · coal
345 kV trunk line
power super-highway
Punta Catalina – Julio Sauri trunk line (752 MW evacuation)
345 kV busbar JULIO SAURI B1 (345 kV yard)
T1 345/138
T2 345/138
138 kV busbar JULIO SAURI B1 (138 kV yard)
T1/T2 power autotransformer banks (345/138 kV)
138 kV line – San Cristóbal Norte (SCN: double circuit L1/L2)
138 kV line – Palenque
138 kV line – Palamara (Santo Domingo ring)
138 kV line – Pizarrete
Nodal legend. One physical estate = 1 substation. Kirchhoff = 2 decoupled nodal electrical busbars separated by the autotransformer impedance. Each voltage yard is a distinct bus: modeling the 345, 230 and 138 kV levels with their transformers, the network sums to exactly 142 nodal buses.